UNION PACIFIC RAILROAD COMPANY

MECHANICAL INSTRUCTIONS

L-14-1

SUBJECT: BOILER DETAILS - SAFE ENDS FOR TUBES AND FLUES.

INSTRUCTIONS:

The following instructions govern the application of safe ends to boiler tubes and flues of all classes of locomotives.

Tubes and flues must be clean and free from scale.

Safe ends shall be cut and tubes and flues shall be reamed to an angle of 35 degrees as shown on drawing, when fire welded.

Safe ends only shall be cut to an angle of 35 degrees, and tubes and flues shall be reamed only enough to remove sharp corners from inner edges as shown on drawing when electric welded in machines on which pressure is used.

APPLICATION OF SAFE ENDS:

(a) Safe ends made by fire weld or electric welding machine in which pressure is used:

Not more than three safe ends on either end are to be applied to 2" and 2-1/4" diameter tubes. No weld to be more than 48" from end of tube.

Not more than two safe ends on either end are to be applied to 3-1/2", 3-3/4" and 4" diameter flues. No weld to be more than 36" from end of flue.

Not more than one safe end on either end is to be applied to 5-3/8" and 5-1/2" diameter fluos. No weld to be more than 24" from end of flue.

Old safe ends on tubes and flues are to be cut off not less than 2" back of first weld and no weld shall be within 4" of end of tube or flue.

(b) Safe ends made on flash butt welding machines used at Cheyenne and Pocatello Shops:

No restrictions as to number of safe ends to be applied to tubes and flues when applied by flash butt welding machine except no weld to be more than 60" from end of tube or flue.